Batch_fft

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Introduction

This graphical user interface has been originally designed to process super-resolution cryo-EM images delivered by the Apollo Camera (Direct Electron) in 8k x 8k format on the JEOL ARM 200 microscope present on the TEM2C platform. Yet, it can be used with any data set. The main objective of **batch_fft** is to provide users files of a sufficiently light format and good contrast, when high-resolution information is not essentially required.

The user selects a folder where super-resolution data are stored and chooses a binning value (2 by default), an image format (TIFF 8-bit by default), and a Gaussian cut-off in Fourier space to increase image contrast.

To find the adequate parameters, one (or a few) file(s) can be opened and inspected, with the possibility to tune both the binning and cut-off-values. Results can be appreciated by inspecting the image and its FFT.

Once these parameters have been determined, they can be entered in the main GUI to process the whole data set.

Installation

- Download batch_fft.py from https://github.com/dchretien35/Batch_FFT, as well as the Logo_TEM2C.png image, and place them in a dedicated folder, named (e.g.) batch_fft.
- It is recommended to create a batch_fft environment:

- Install miniconda3 (if not yet present in your system):

- Open a terminal, and at the command line run:
`conda create --name batch_fft python=3.13` (or your Python version)

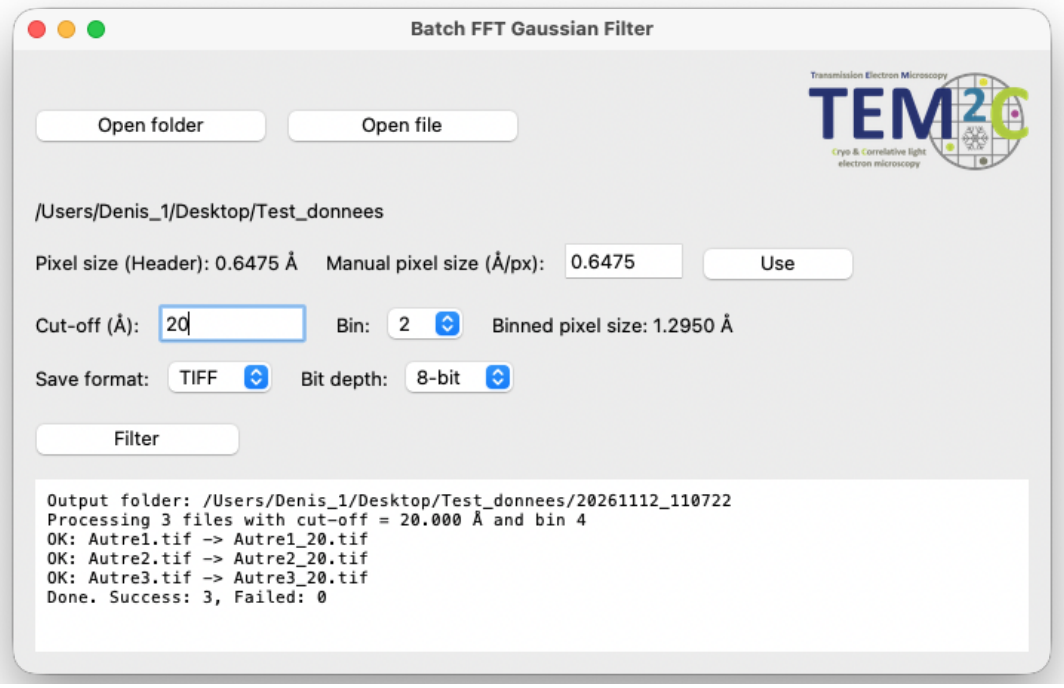
- Then enter: `conda activate batch_fft`

- To quit the environment, use: `conda deactivate`

- Python: Python 3.9+ (3.10+ recommended).
- Needs Tkinter available with the Python build (standard on most macOS/Windows installers; on some Linux distros: run 'install python3-tk').
- Packages for full feature set (projections and gallery): numpy, mrcfile, Pillow.
- One-line install (inside your desired env): run '`pip install numpy mrcfile Pillow`

Procedure

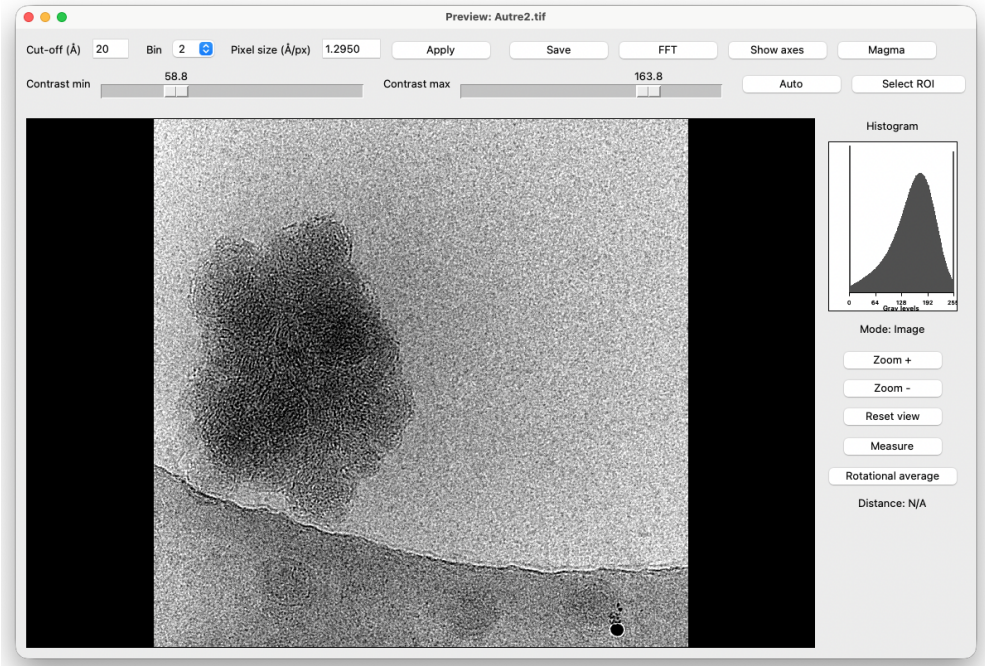
- Once in your environment, type at the terminal: `python3 batch_fft.py`



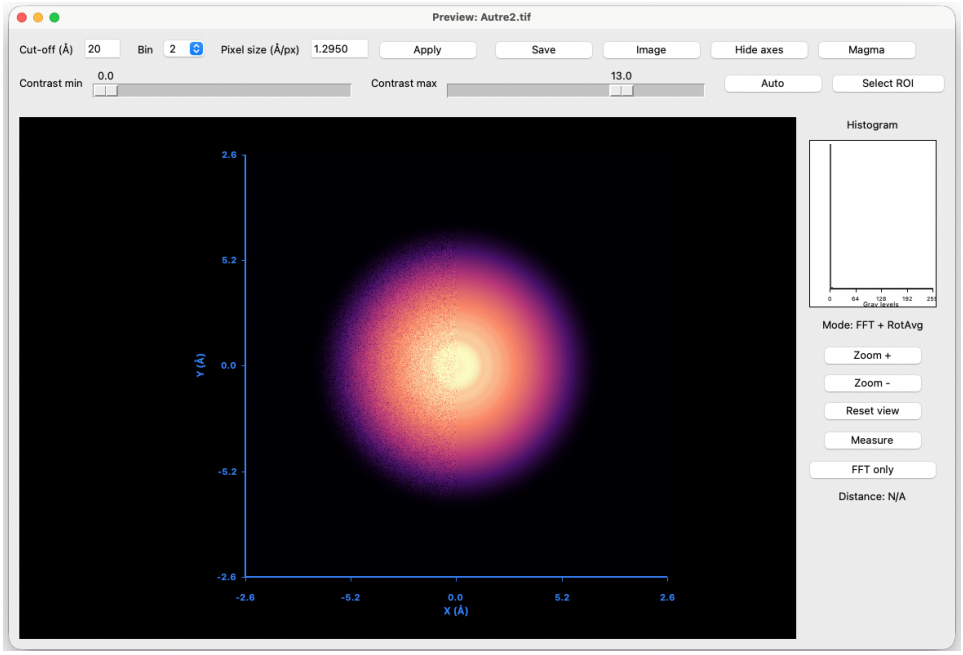
- **Open folder**: allows you to choose the folder where the cryo-EM images that need to be processed are present. The pixel-size is read from the file header.

  Note that other cryo-EM images than the one coming from the Apollo can be processed, even those lacking a header. In that case, the pixel size can be entered manually (**Manual pixel size (Å/px):**, then click on **Use**).

- **Cut-off (Å)**: Spatial periodicity cut-off used to filter the images. A low cut-off may provide noisy images, while a too-high cut-off may blur them. A reasonable cut-off can be estimated by inspecting a few files using the **Open file** menu (see below).
- **Bin**: Binning value (1: no binning, up to 8). Super-resolution data coming from the Apollo are 8192 x 8192 pixel size (134.2 Mo), which takes a large amount of disc space. A binning by 2 (default) preserves image quality and the header information if saved in TIFF format.
- **Save format**: This menu allows saving file in TIFF (16-bit or 8-bit), or PNG and JPEG format. While these latter formats can save a large amount of disc space, they contain no header. Hence, the pixel size must be noted for further image analysis.
- **Filter**: Used to process files with the given parameters. A subfolder is created which contains the name **date_time**. Files keep their original name with the added cut-off used, e.g., filename_20 for a 20 Å cut-off.
- **Information window**: Provides information such as files processed.
- **Open file**: This menu is used to estimate adequate parameters before filtering the whole data set.



- **Cut-off** and **Bin** can be used to find the right values by inspecting visually image quality.
- **Pixel-size (Å/px)**: Recovered form the GUI, and updated if the Bin value is modified. In case of images with no header, the pixel size can be entered here (take into account the binning, or tun it to 1 before entering this value).
- **Apply**: To apply changes in cut-off and binning values (‘Enter key’ works for cut-off values).
- **Save**: The displayed image can be saved in an individual folder (same rules as previously), taking into account any modification such as contrast adjustment.
- **FFT**: Toggles between imaging and FFT modes.



In FFT mode, two parameters can be estimated. First the size of the filter, which depends on both the cut-off and binning values, and second the image quality and approximate resolution by looking at the Thon rings in the FFT.

- **Show/Hide axes**: Displays or hides (default) axes on the image or the FFT (as exampled above).
- **Magma/Grey**: toggles the FFT display between magma colors and grey levels.
- **Contrast sliders**: allows contrast tuning of the displayed image/FFT. This has no impact on the files that will be processed, which is true for other functions (e.g., Auto or ROI contrast adjustment).
- **Auto**: performs an automatic contrast adjustment of the whole image/FFT.
- **Select ROI**: allows contrast adjustment of the whole image/FFT based on the contrast of a selected region. This can also be performed using the right-click mouse.
- **Histogram**: displays the grey levels (useful in imaging mode).
- **Zoom +** and **Zoom –** : Allows zooming on the image/FFT. This can also be performed using the middle mouse wheel, but the zoom buttons keep the image centered. Note that the image/FFT can be moved with the left-click mouse button.
- **Measure**: draws a line on the image and reports the distance in nm (image) or Å (FFT). This latter measurement can be useful to estimate the image resolution (last Thon rings), or a repetition in a crystalline or helical specimen.
- **Rotational average**: performs a rotational average of the FFT to reinforce Thon ring contrast. It is then juxtaposed on the right part of the FFT, such as displayed above.